

*“The difference between theory and
practice
is larger in practice than in theory.”*

— Jan van de Snepscheut, Computer Scientist

Abstract

This paper provides computational validation of the **Cognitive Integrity Framework (CIF)** introduced in Part 1. We implement the complete defense suite—cognitive firewalls, belief sandboxes, trust calculus, and Byzantine-tolerant consensus—and evaluate performance across production multiagent architectures.

Contributions

- ▶ **Attack Corpus:** 950 cognitive attacks across four categories (prompt injection, trust exploitation, belief manipulation, coordination attacks)
- ▶ **Cross-Architecture Validation:** Evaluation across Claude Code, AutoGPT, CrewAI, LangGraph, MetaGPT, and Camel
- ▶ **Detection Performance:** 94% detection rate with layered defenses; 20-25% latency overhead
- ▶ **Statistical Analysis:** Significance testing with large effect sizes (Cohen's $d > 0.8$), ablation studies, scalability benchmarks

Key Findings